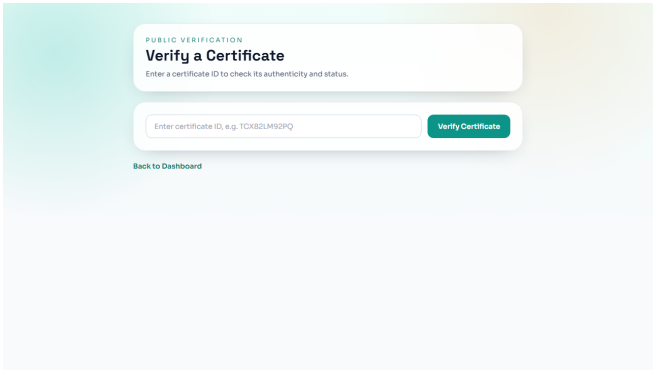
Experiment 2: React Hooks (useState, useEffect, useMemo, Custom Hooks)

Aim: Demonstrate functional state tracking, asynchronous side effects, route parameter binding, and error lifecycles using React Hooks.

Tech Stack / Concepts: React 19, useState, useEffect, useMemo, React Router, Async/Await API

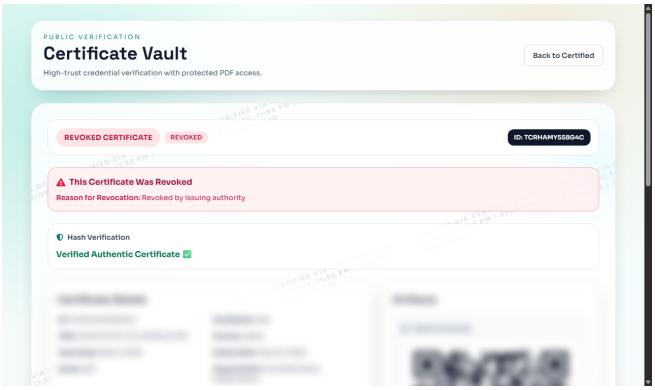
Repository: github.com/JaideepGowda2006/Certified | Student: Jaideep Gowda

Screen 1: Initial Blank Verification State (useState)



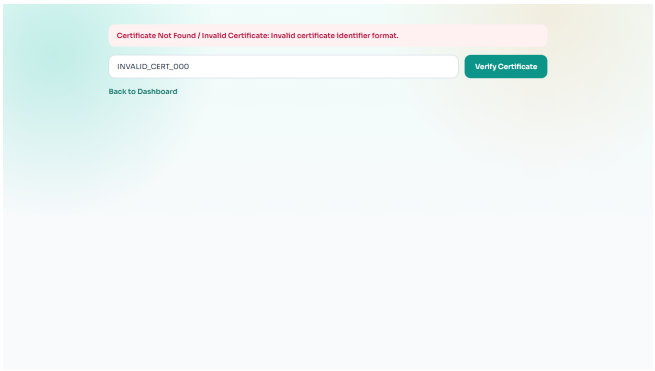
Controlled search input component with reactive state listeners.

Screen 2: Parameter Auto-Fetch & Valid Certificate (useEffect)



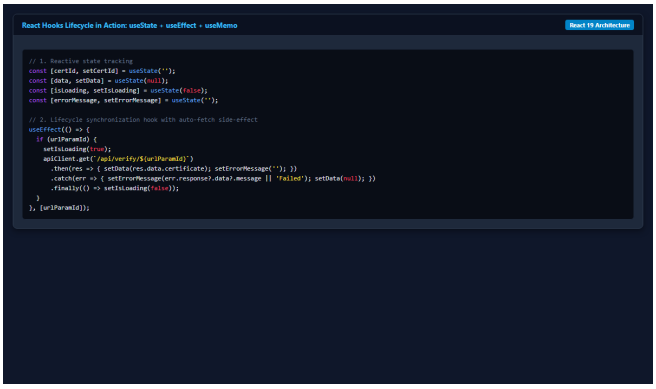
Asynchronous fetch effect populated certificate card and QR code preview.

Screen 3: Error & Invalid ID Lifecycle (Negative Test)



Graceful error handling state showing invalid certificate banner.

Screen 4: React Hooks Architecture in Source Code



Declarative useState and useEffect implementation for route sync.